

⚡ 30-Day Co-op AI Playbook & Executive

Reference

From Conversational Chatbots to Bounded, Auditable IT Workflows

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2026 CREA IT Conference • Lakewood, CO

Core Thesis: Taking AI to the next level is not about prompting chatbots; it is about building bounded, auditable IT workflows where AI retrieves, analyzes, and prepares work under strict human governance—while keeping the Operational Technology (OT) air gap absolute.

1. The Sacred Boundary: OT vs. IT Critical Isolation

[OPERATIONAL TECHNOLOGY (OT)] [ENTERPRISE IT NETWORK] SCADA • Substations • Breakers CIS/Billing • Ticketing • Docs | |
|
OPERATIONAL CAPABILITY • Zero control access • Bounded Agentic Pipelines • NERC CIP ESP contained • Local Inference (Ollama/vLLM)
• Relays unidirectional • Mandatory Human-in-the-Loop Gate

Rule 1: AI enhances Enterprise IT productivity; AI never touches the breaker loop. Keeping AI workloads strictly on Enterprise IT subnets prevents expanding the NERC CIP Electronic Security Perimeter (ESP).

2. The 4-Level AI Maturity Ladder

Level	Definition & Mechanics	Utility Cooperative Application
L4: OPERATE FUTURE	Autonomous scheduled pipelines & monitoring	Night-time firewall drift verification, automated log anomaly detection.
L3: BOUNDED EXECUTE TODAY'S HERO	Multi-step agent tools + Human Approval Gates	IT ticket parsing → SOP cross-reference → Staged PowerShell remediation.
L2: GROUND NEXT STEP	Retrieval-Augmented Generation (RAG)	Semantic search across 20 yrs of scanned vendor PDFs (SEL-751) & audio briefings.
L1: ASSIST BASELINE	One-off chat prompts (ChatGPT, base Copilot)	General email drafting, text summarizing (unaware of internal co-op data).

3. The Guardrail Bounded Agent Operating Loop

1. Human Intent & SoT Creation
Sysadmin defines goals & constraints. Orchestrator consults ground-truth Markdown Source of Truth (SoT).

2. Verification Sandbox
Agent lints scripts, tests SQL queries, and checks diffs in isolation before staging any change package.

3. HUMAN APPROVAL GATE
MANDATORY: Execution halts. Sysadmin reviews verified package and clicks 'Approve & Execute'.

4. Production Apply + Git Rollback
Change applied, logged in immutable audit ledger, and backed by instantaneous Git version-control recovery.

4. The AI Security Contract (6 Governance Guardrails)

**1. Data Classification**

Tag and filter data prior to ingestion (Public vs. Restricted) to prevent member PII leakage.

**2. Least Privilege Tooling**

Default read-only tool permissions; write actions strictly scoped to isolated endpoints.

**3. Verification Sandboxing**

All scripts syntax-checked and linted by secondary critic models prior to human review.

**4. Human Approval Gates**

High-risk system changes require explicit sysadmin sign-off with clear diff previews.

**5. Immutable Audit Logs**

Every user prompt, tool call, and script execution recorded in a timestamped compliance ledger.

**6. One-Click Git Rollback**

Version control (Git) backing every file modification to ensure instant state restoration.

5. On-Premises Hardware Bill of Materials (BoM)

Tier	Hardware Specifications	Model Sizing	CAPEX	Target Utility IT Use Case
Entry Host	Apple Mac Studio (64–128GB Unified)	32B–70B Quantized	\$2.5k–\$4.5k	Private document search, sensitive script drafting, zero cloud leak.
Dedicated Node	NVIDIA DGX Spark / Compact Node	70B Swarms	~\$4.7k	Departmental inference server, continuous log RAG, internal API.
High-Throughput	Dual RTX 4090 / RTX 6000 Ada	70B+ FP16 High-Conc	\$8k–\$18k	High-volume OCR document pipelines, multi-user co-op portal.

*Operating costs: Local inference power draw is ~\$15–\$45/month, compared to \$35–\$95/mo for metered cloud APIs.

6. The 4-Week Action Sprint (Monday Morning Checklist)

WEEK 1: SELECT & CLASSIFY — Pick ONE high-friction, document-heavy IT workflow (e.g. legacy vendor manuals or VPN troubleshooting). Classify data sensitivity (Public vs Internal).

WEEK 2: GROUND & CLEAN — Convert PDFs to clean Markdown; ingest into a grounded tool (Gemini Notebook / Ollama). Verify cited retrieval accuracy and test 15-second audio briefings.

WEEK 3: PILOT BOUNDED LOOP — Deploy a Level 3 agent workflow with a mandatory **Human Approval Gate**. The agent stages PowerShell commands or ticket responses; the admin signs off.

WEEK 4: MEASURE & EVALUATE — Measure technician hours saved, audit error rate, calculate ROI, and present findings to GM/Board before purchasing new hardware.